

IMO 2026 — Problem 5

Solution by DeepSeek V4 Pro

2026-07-21

Source: results/deepseek-v4-pro/problem-05/current.md — autonomous run, no human intervention; solution as written by the model.

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2 + f(y)^2}{2}} \geq \frac{f(x) + y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Solution document

Status

solved

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that

$$\sqrt{\frac{x^2 + f(y)^2}{2}} \geq \frac{f(x) + y}{2} \geq \sqrt{xf(y)}$$

for every $x, y \in \mathbb{R}_{>0}$.

Approaches tried

1. Numerical brute-force — only $f(x) = x + c$ ($c \geq 0$) survives.
2. Substituting $x = f(y)$ forces the functional equation $f(f(x)) = 2f(x) - x$.
3. Reformulating with $g(x) = f(x) - x$ yields two algebraic identities for the squared-inequality remainders A, B .
4. From $|A - B| \leq A + B$ we obtain a universal inequality (7).
5. The functional equation implies forward orbits are arithmetic progressions with common difference $g(x)$. If $g(x) < 0$ the orbit leaves $\mathbb{R}_{>0}$ – contradiction. Hence $g \geq 0$.
6. Using arithmetic progressions together with (7) shows that g attains at most one positive value.
7. A backward-induction lemma gives a point $u \in (0, c]$ with $g(u) = c$ (if $c > 0$ is a value).

8. A covering argument (intervals around the forward orbit) forces the zero-set $S_0 = \emptyset$, hence g is constant c .
9. Verification that $f(x) = x + c$ works for any $c \geq 0$.

Current best

Answer: $f(x) = x + c$ for a constant $c \geq 0$.

Full proof

Let $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ satisfy

$$\sqrt{\frac{x^2 + f(y)^2}{2}} \geq \frac{f(x) + y}{2} \geq \sqrt{xf(y)} \quad \forall x, y > 0. \quad (1)$$

All quantities are positive, so squaring preserves the inequalities:

$$2x^2 + 2f(y)^2 \geq (f(x) + y)^2 \geq 4xf(y) \quad \forall x, y > 0. \quad (2)$$

1. A functional equation

Put $x = f(y)$ (which is > 0) in (2). The leftmost term becomes $2f(y)^2 + 2f(y)^2 = 4f(y)^2$ and the rightmost becomes $4f(y)f(y) = 4f(y)^2$. Hence the chain collapses to equality:

$$(f(f(y)) + y)^2 = 4f(y)^2.$$

Taking square roots (all terms are positive) yields

$$f(f(y)) + y = 2f(y) \quad \implies \quad \boxed{f(f(x)) = 2f(x) - x \quad \forall x > 0.} \quad (3)$$

Define

$$g(x) = f(x) - x \quad (x > 0).$$

Then $g(x) > -x$ and (3) rewrites as

$$x + g(x) + g(x + g(x)) = 2x + 2g(x) - x \implies \boxed{g(x + g(x)) = g(x) \quad \forall x > 0.} \quad (4)$$

2. Positivity of g

Lemma 0. $g(x) \geq 0$ for all $x > 0$.

Proof. Fix $x > 0$ and define the forward orbit $x_0 = x$, $x_{n+1} = f(x_n)$ for $n \geq 0$. Using (3),

$$x_{n+2} = f(x_{n+1}) = 2f(x_n) - x_n = 2x_{n+1} - x_n.$$

The recurrence $x_{n+2} - 2x_{n+1} + x_n = 0$ has the unique solution $x_n = A + Bn$ with A, B determined by the initial conditions: $x_0 = x$, $x_1 = f(x) = x + g(x)$. Hence $A = x$, $B = g(x)$, and

$$x_n = x + ng(x) \quad (n \geq 0).$$

From (4) we have $g(x_n) = g(x)$ for all $n \geq 0$. Now, if $g(x) < 0$, then x_n becomes non-positive for sufficiently large n , contradicting that every x_n is in the domain $\mathbb{R}_{>0}$ (because $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$). Therefore $g(x) \geq 0$.

Thus g takes only non-negative values.

3. Algebraic reformulation

Write $a = g(x)$, $b = g(y)$; then $f(x) = x + a$, $f(y) = y + b$. The two inequalities in (2) are

$$\begin{aligned} A(x, y) &:= 2x^2 + 2(y + b)^2 - (x + y + a)^2 \geq 0, \\ B(x, y) &:= (x + y + a)^2 - 4x(y + b) \geq 0. \end{aligned}$$

Expanding:

$$\begin{aligned} A(x, y) &= (x - y)^2 - a^2 - 2a(x + y) + 4by + 2b^2, \\ B(x, y) &= (x - y)^2 + a^2 + 2a(x + y) - 4bx. \end{aligned}$$

Adding and subtracting gives two fundamental identities

$$A(x, y) + B(x, y) = 2(x - y - b)^2 = 2(x - f(y))^2, \quad (5)$$

$$A(x, y) - B(x, y) = -2(a - b)(a + b + 2x + 2y). \quad (6)$$

Because $A, B \geq 0$, we have $|A - B| \leq A + B$ (the absolute difference of two non-negative numbers never exceeds their sum). Substituting (5) and (6) and cancelling a factor 2 yields

$$\boxed{|g(x) - g(y)| (g(x) + g(y) + 2x + 2y) \leq (x - y - g(y))^2 \quad \forall x, y > 0.} \quad (7)$$

Swapping x and y gives the symmetric inequality

$$|g(x) - g(y)| (g(x) + g(y) + 2x + 2y) \leq (y - x - g(x))^2 \quad \forall x, y > 0. \quad (8)$$

4. At most one positive value of g

Let $V = \{g(x) : x > 0\}$. Suppose V contains two distinct positive numbers $c_1 > c_2 > 0$. Choose $u, v > 0$ with $g(u) = c_1$, $g(v) = c_2$. From the functional equation (4) we obtain

$$g(u + nc_1) = c_1, \quad g(v + mc_2) = c_2 \quad \forall n, m \in \mathbb{N}_0.$$

Insert $x = u + nc_1$, $y = v + mc_2$ into (7). With $a = c_1$, $b = c_2$,

$$(c_1 - c_2)(c_1 + c_2 + 2u + 2v + 2nc_1 + 2mc_2) \leq (u - v - c_2 + nc_1 - mc_2)^2. \quad (9)$$

We show that for suitably chosen $n, m \rightarrow \infty$ the inequality fails.

Rational ratio. If $c_1/c_2 \in \mathbb{Q}$, write $c_1/c_2 = p/q$ with $p, q \in \mathbb{N}$ coprime. Then $pc_2 = qc_1$. Take $n = qt$, $m = pt$ for $t \in \mathbb{N}$. Then $nc_1 - mc_2 = 0$, so the right-hand side of (9) is the constant $(u - v - c_2)^2$, while the left-hand side grows linearly with t (coefficient $2(c_1 - c_2)(qc_1 + pc_2) > 0$). For large t the inequality is violated – contradiction.

Irrational ratio. If $c_1/c_2 \notin \mathbb{Q}$, the fractional parts $\{n(c_1/c_2)\}$ are dense in $[0, 1]$ (a classical fact: for irrational α , the sequence $n\alpha \bmod 1$ is dense). Consequently the set $\{nc_1 - mc_2 : n, m \in \mathbb{N}_0\}$ is dense in $\mathbb{R}$. Hence we can choose sequences $(n_k, m_k) \rightarrow (\infty, \infty)$ with $n_k c_1 - m_k c_2 \rightarrow -(u - v - c_2)$. Then the right-hand side of (9) tends to 0 while the left-hand side tends to $+\infty$, again a contradiction.

Thus V contains **at most one** positive number. Because $g \geq 0$, the only possible values are 0 and at most one $c > 0$. Consequently, either

- (i) $g \equiv 0$ (constant), which gives $f(x) = x$, a solution; or
- (ii) there is $c > 0$ such that $V \subseteq \{0, c\}$ with both values attainable.

We now exclude case (ii). Assume $c > 0$ and both 0 and c occur. Set

$$S_0 = \{x > 0 : g(x) = 0\}, \quad S_c = \{x > 0 : g(x) = c\},$$

both non-empty. (Because $V \subseteq \{0, c\}$, every $x > 0$ belongs to $S_0 \cup S_c$.)

5. Inequalities between the two level sets

Take $x \in S_0$, $y \in S_c$ ($a = 0$, $b = c$). The formulas for A and B become

$$\begin{aligned} A(x, y) &= (x - y)^2 + 4cx + 2c^2 \geq 0 \quad (\text{always true}), \\ B(x, y) &= (x - y)^2 - 4cx \geq 0 \implies (x - y)^2 \geq 4cx, \quad (10) \\ A(y, x) &= (x - y)^2 - c^2 - 2c(x + y) \geq 0 \implies (x - y)^2 \geq c^2 + 2c(x + y). \quad (11) \end{aligned}$$

From (11) and $x, y > 0$ we obtain $(x - y)^2 > c^2$, hence

$$|x - y| > c \quad \forall x \in S_0, y \in S_c. \quad (12)$$

6. Existence of a small element in S_c

Lemma 1. There exists $u \in (0, c]$ with $g(u) = c$.

Proof. Pick any $y_0 \in S_c$. If $y_0 \leq c$ we are done. Suppose $y_0 > c$ and set $z = y_0 - c > 0$. Apply inequality (7) with $x = y_0$, $y = z$. We have $a = g(y_0) = c$, and $b = g(z)$ belongs to $\{0, c\}$ (all values are in $V \subseteq \{0, c\}$).

Compute:

$$\begin{aligned} \text{LHS} &= |c - b|(c + b + 2y_0 + 2z) = |c - b|(c + b + 2y_0 + 2(y_0 - c)) = |c - b|(b - c + 4y_0), \\ \text{RHS} &= (y_0 - z - b)^2 = (y_0 - (y_0 - c) - b)^2 = (c - b)^2. \end{aligned}$$

Let $d = b - c$. Then $\text{LHS} = |d|(d + 4y_0)$, $\text{RHS} = d^2$.

- If $d > 0$: $d(d + 4y_0) \leq d^2 \implies 4y_0d \leq 0$, impossible.
- If $d = 0$: inequality holds as $0 \leq 0$.
- If $d < 0$: $|d| = -d$, giving $(-d)(d + 4y_0) \leq d^2 \implies -d^2 - 4y_0d \leq d^2 \implies -4y_0d \leq 2d^2$. Since $d < 0$, divide by $-2d > 0$: $2y_0 \leq -d = |d|$.

Thus $|d| \geq 2y_0$. But $b = g(z) \in \{0, c\}$, so $|d| = |b - c|$ is either 0 (if $b = c$) or c (if $b = 0$).

If $b = 0$, then $|d| = c$, giving $c \geq 2y_0$, i.e. $y_0 \leq c/2$, contradicting $y_0 > c$. Therefore $b \neq 0$, so $b = c$; hence $g(z) = c$.

Thus $z = y_0 - c \in S_c$. If $z > c$, repeat the argument with y_0 replaced by z . After $\lfloor y_0/c \rfloor$ steps we reach an element $u \in (0, c]$ belonging to S_c .

7. The forward orbit and a covering argument

From Lemma 1 we have $u \in (0, c]$ with $g(u) = c$. By (4),

$$g(u + nc) = c \quad \forall n \in \mathbb{N}_0.$$

Thus the arithmetic progression

$$P = \{u + nc : n \geq 0\} \subseteq S_c.$$

Consider the closed intervals

$$I_n = [u + nc - c, u + nc + c] \quad (n \geq 0).$$

Their union covers all positive real numbers. Indeed, $I_0 = [u - c, u + c]$; because $0 < u \leq c$, the left endpoint is ≤ 0 , so I_0 contains $(0, u + c]$. $I_1 = [u, u + 2c]$, $I_2 = [u + c, u + 3c]$, ... Since I_n and I_{n+1} overlap (their intersection is $[u + nc, u + (n + 1)c]$), we have

$$\bigcup_{n \geq 0} I_n = (0, \infty).$$

Now take any $x > 0$. There exists $n \geq 0$ such that $x \in I_n$, i.e.

$$|x - (u + nc)| \leq c.$$

If $x \in S_0$, then by (12) we must have $|x - (u + nc)| > c$ (because $u + nc \in S_c$). This is impossible. Hence $S_0 = \emptyset$.

8. Conclusion

Since $S_0 = \emptyset$, the function g never takes the value 0. Together with §4 this forces $g(x) \equiv c$ for the positive constant c . Thus

$$f(x) = x + c \quad (c > 0).$$

The earlier case $g \equiv 0$ gives $f(x) = x$, which corresponds to $c = 0$. Both are covered by the single formula

$$\boxed{f(x) = x + c \quad \text{for some constant } c \geq 0.}$$

Verification. For $f(x) = x + c$ ($c \geq 0$) the original inequalities become

$$\sqrt{\frac{x^2 + (y + c)^2}{2}} \geq \frac{x + y + c}{2} \geq \sqrt{x(y + c)},$$

which are exactly the Quadratic-Mean–Arithmetic-Mean inequality and the Arithmetic-Mean–Geometric-Mean inequality applied to the two positive numbers x and $y + c$. Both inequalities are true for all $x, y > 0$.

Therefore the set of all solutions is $\{x \mapsto x + c \mid c \geq 0\}$.